

Learning seminar on Number Theory

Cornell, Spring Semester 2026

Hodge-Tate decomposition and p -divisible groups

Overview

Given a ℓ -adic Galois representation $\rho : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow V$ coming from geometry, *i.e.*, appears as subquotient of some étale cohomology group, it is more complicate(=rich) to look at the restriction to the decomposition group at $p = \ell$ than any other primes. In the spirits of comparisons between different cohomologies as in Hodge theory, p -adic Hodge theory also connects various cohomology theories with arithmetic flavor, and is more about using other cohomology to understand local behavior of Galois representations.

Theorem 1 (Hodge-Tate decomposition). *Let $K/\mathbb{Q}_p$ be a finite extension, and let $\mathbb{C}_p$ be a completion of an algebraic closure $\overline{K}$ of K . Let X/K be a smooth proper variety. Then there exists a Galois equivariant decomposition*

$$H_{\text{ét}}^n(X_{\overline{K}}, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} \mathbb{C}_p \simeq \bigoplus_{i+j=n} H^i(X, \Omega_{X/K}^j) \otimes_K \mathbb{C}_p(-j), \quad (1)$$

where $C_p(-j)$ denotes the $(-j)$ -th Tate twist of C_p . This isomorphism is functorial in X . In particular, it respects the natural graded algebra structures on either side as n varies.

The standard arguments of Kummer exact sequence and duality of abelian varieties shows the first cohomology group is isomorphic to the dual of Tate module when X is an abelian variety. Therefore, one can read off the Galois action from the right in (1).

In this seminar we plan to learn the proof of this theorem using p -divisible groups [Tat67], which is an intermediate between abelian varieties and their Tate modules and capture the local geometric information better. Another modern reference p -divisible groups are [Hon]. For some motivation notes, see [Con], [Ayo].

Talks

Talk 1: Introduction and Logistics

We try to motivate everyone in this topics and settle logistics stuff.

Talk 2: Affine group schemes and affine subgroups

Define group schemes and give examples; show their relation between cocommutative Hopf algebras. Briefly recall functor points of view and explain it for some group schemes. Define affine subgroup and discuss Hopf ideal. State and prove 2.5-2.8 in [Mor15].

Talk 3: Affine group schemes I

Define morphisms and the category of affine group schemes. Introduce the subscheme of n torsion group as much as possible. If time permits, discuss the n torsion points of elliptic curves as a group scheme.

Talk 4: Finite flat group scheme

Define injection, surjection and exactness, state and prove that taking R -points is left exact. Define finite flat group schemes, state and prove Prop 3.5 in [Mor15], except (1). State the relation of finite étale group schemes and Galois modules in char 0. Mention n torsion points of elliptic curve as flat groups schemes (they are always flat but when are they étale?), if time permits.

Talk 5: p -divisible groups

Define p -divisible group, also mention Grothendieck's equivalent definition (as fppf sheaves), and give examples. Define Tate modules and comodules of p -divisible groups, and show that they are Galois representations. Define tangent space and state the main theorem.

Talk 6: Formal group schemes and Serre-Tate

Define formal groups, show that they are group objects in the category of formal schemes. Prove Serre-Tate equivalence, which is Prop 4.12 in [Mor15].

Talk 7: Connected-étale sequence

Prove connected-étale sequence as in Prop 5.7 in [Mor15] and state their universal properties. Also state (and prove if time permits) Lemma 5.8. State these results for p -divisible groups.

Talk 8: R -points

Define R -points of p -divisible groups (mention $R = \mathcal{O}_{\mathbb{C}_p}$ usually). Go through everything between 6.3-6.5 in [Mor15].

Talk 9: Cartier duality and classification

Discuss Cartier duality as much as possible. State (not necessarily prove) a classification of finite flat group schemes over an algebraically closed field, for example in [Dem72]. Start the proof of the main theory if time permits.

Talk 10: Proof

Prove the main theorem I.

Talk 11: Proof

Prove the main theorem II.

Talk 12: Proof

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References

- [Ayo] <https://ayoucis.wordpress.com/2015/08/10/p-divisible-groups-formal-groups-and-the-serre-tate-theorem>.
- [Con] Brian Conrad. <https://virtualmath1.stanford.edu/~conrad/mordellsem/Notes/L09.pdf>. Lecture notes (Lecture 09) from the course on Mordell's conjecture and related topics, Stanford University.
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- [Hon] Serin Hong. Notes on p -adic hodge theory.
- [Mor15] Matthew Morrow. p -divisible groups, 2015.
- [Tat67] J. T. Tate. p -divisible groups. In *Proc. Conf. Local Fields (Driebergen, 1966)*, pages 158–183. Springer, Berlin-New York, 1967.